



MARTIN-LUTHER-UNIVERSITÄT
HALLE-WITTENBERG

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Chair of Empirical Microeconomics

ADVANCED MICROECONOMICS

Summary & Questions

Winter Term 2025/2026

What was this course about?

- **Goal:** Understand economic decision making (focus: consumption choices)
- **Core idea:** Reasoning from **preferences** to **individual choices** to **welfare/markets**
- We concentrated on a **normative theory** (rational choice/standard neoclassic model) ...
- ... we expanded it to situations under risk ...
- ... and started to learn when/why it **fails** descriptively (information frictions, behavioral biases)

Understand theory → see limits → understand actual behavior

Outline

1) Consumer Theory

- 1.1) Preferences and Utility (L1)
- 1.2) Utility Maximization (L2)
- 1.3) Expenditure Minimization (L3)
- 1.4) Demand Functions and Duality (L4+L5)

2) General Equilibrium

- 2.1) Exchange Economy (L6)
- 2.2) Welfare Economics and General Equilibrium (L7)

3) Decisions under Risk and Uncertainty

- 3.1) Expected Utility (L8)
- 3.2) Risk Preferences (L9)
- 3.3) Insurances (L10)
- 3.4) Implicit Markets for Risk and the Value of a Statistical Life (11)

4) Behavioral Economics (L12)

Overview

1 Consumer Theory

Question: Homothetic Preferences

Question: Slutsky Decomposition

2 General Equilibrium

3 Decision under risk and uncertainty

Question: Certainty Equivalent

4 Behavioral Economics

5 Outlook

6 Hints for the Exam

Consumer Theory

Approach

- Standard theory assumes that consumers make their decisions so as to make themselves as well off as possible (i.e., maximize their utility)
- Under **specific assumptions**, we can express individuals' utility as mathematical functions that represent underlying preferences.
- Two dual approaches to get individual decisions:
 - Constrained utility maximization
 - Expenditure minimization
- Both approaches results in demand functions, which allow to analyze changes in demand when prices change (income and substitution effect)

Preferences: the primitives

We start with preferences $\succeq$ over bundles.

Standard assumptions (why we like them):

- **Completeness + transitivity + continuity** $\Rightarrow$ consistent choice & utility representation
- **Monotonicity (non-satiation)** $\Rightarrow$ budget exhaustion
- **Convexity** $\Rightarrow$ love for diversification, interior solutions

Everything else in the consumer model is built on these assumptions.

Utility is ordinal: monotone transformations

Utility is a **representation** of preferences, not “happiness”.

Monotone transformation: If $u(x)$ represents $\succeq$, then so does

$$\tilde{u}(x) = f(u(x)) \quad \text{for any strictly increasing } f(\cdot).$$

- Allows for behaviorally meaningful statements about choice bundles/demand but **not** utility **levels**.

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Homothetic Preferences: Intuition

- Homotheticity is another possible property of preferences but is not one of the key axioms
- **Why?** It does not guarantee existence of demand, uniqueness of solutions or well-behaved optimization and is thus not essential.
- It still is very intuitive and most preference relations we discuss have this property

Homotheticity

Preferences are **homothetic** if scaling consumption does not change rankings. If bundle x is preferred to bundle y , then a scaled-up version of x is also preferred, i.e. if for all x, y and all $t > 0$:

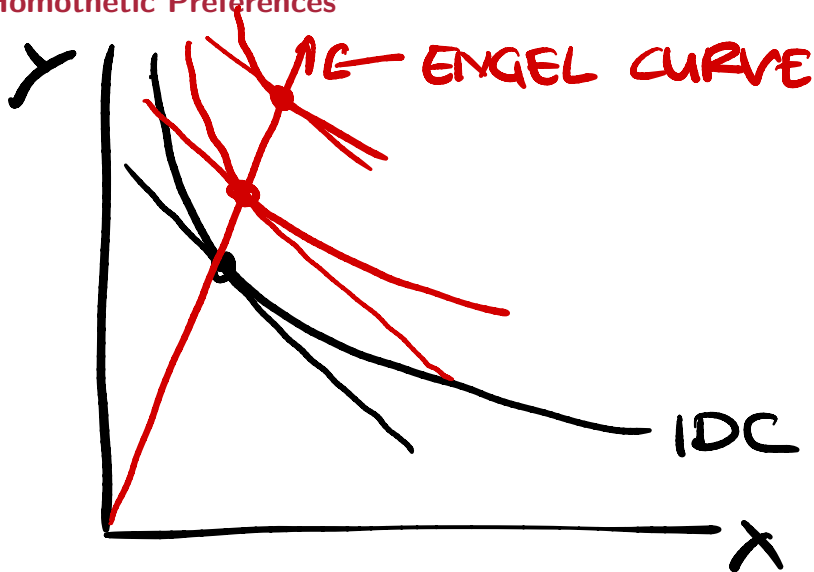
$$x \succ y \Rightarrow tx \succ ty$$

Homothetic Preferences

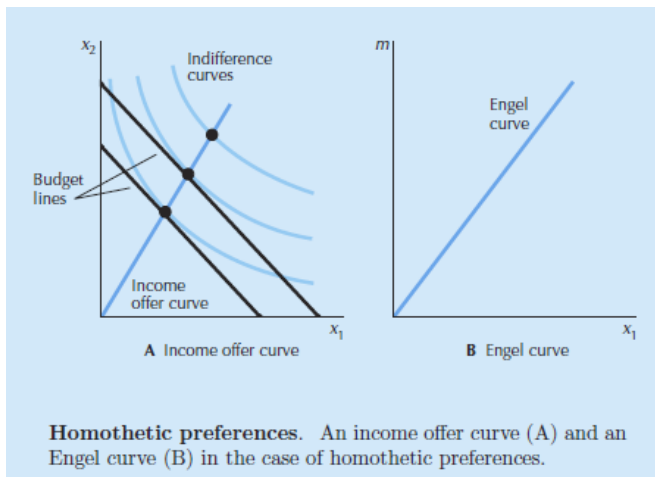
Key intuition:

- Only the *direction* of the bundle matters, not its scale
- Indifference curves are **radial expansions** of each other
 - Indifference curves have the **same shape**
 - Higher indifference curves are scaled-up versions of lower ones
 - Rays from the origin intersect each indifference curve exactly once
 - Income expansion path (Engel curve) is straight line

Homothetic Preferences



Homothetic Preferences



Source: Varian (2020), page 102

Homotheticity and Monotone Transformations

Crucial distinction:

- Homotheticity is a property of **preferences**
- Homogeneity is a property of a **utility function**:
 - A utility function $u(x)$ is homogeneous of degree k if, for all $t > 0$
$$u(tx) = t^k u(x)$$
 - If you scale all goods by a factor t , utility scales by a factor t^k , i.e. if $k = 1$ utility doubles when consumption doubles
- If preferences are homothetic, there exists a utility representation that is homogeneous of degree 1

Why this is (implicitly) important?

Reminder: Utility functions are ordinal $\rightarrow$ monotonous transformation $\tilde{u}(x) = f(u(x))$ represents the **same preferences**.

Key result:

- After a monotone transformation, homogeneity may disappear, **homotheticity never disappears**

Consider the Cobb-Douglas utility function:

$$u(x_1, x_2) = x_1^{1/2} x_2^{1/2}$$

- Homogeneous of degree 1 and represents homothetic preferences

Apply a monotone transformation:

$$\tilde{u}(x_1, x_2) = \ln(x_1^{1/2} x_2^{1/2})$$

- Preferences unchanged (still homothetic), indifference curves unchanged utility is **no longer homogeneous**

Consumer problem: utility maximization

$$\max_x u(x) \quad \text{s.t.} \quad p \cdot x \leq I$$

Graphical intuition:

- Best affordable indifference curve
- Tangency (interior solution): $MRS_{12}(x^*) = \frac{p_1}{p_2} \rightarrow$ “*Psychic trade-off equals market trade-off.*”
- Solution yields **Marshallian demand** $d_x = x(p, I)$ and the **indirect utility function** $V(p, I) = u(d_x(p, I))$, i.e. the maximum attainable utility at (p, I)

Consumer Problem: Duality

Two equivalent ways to describe the same choice problem:

Primal: maximize utility given budget

Dual: minimize expenditure for a target utility level

$$\min_x p \cdot x \quad \text{s.t.} \quad u(x) \geq \bar{u}$$

- Outputs:
 - **Hicksian demand** $h(p, \bar{u})$ (compensated choices)
 - **Expenditure function** $E(p, \bar{u})$ (money metric for utility)
- Why duality matters: Expenditure function translates utility changes into **euros** / money metric.
- Policy language, e.g. **Compensating variation (CV)** as the money needed *after* a change to restore old utility

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Critical concepts - Slutsky Equation

Key question of consumer theory: What happens to demand if prices change?

Approach which follows from duality: The total change in demand d_x (Marshallian) has two components:

- **Substitution effect (SE):** hold utility constant, i.e. isolated with Hicksian demand h_x
- **Income effect (IE):** change in purchasing power

$$\underbrace{\frac{\partial d_x}{\partial p_x}}_{\text{total}} = \frac{\partial h_x}{\partial p_x} - \frac{\partial E}{\partial p_x} \frac{\partial d_x}{\partial I} = \underbrace{\frac{\partial h_x}{\partial p_x}}_{\text{SE}} - \underbrace{x \frac{\partial d_x}{\partial I}}_{\text{IE}}$$

Slutsky vs. Hicks Decomposition

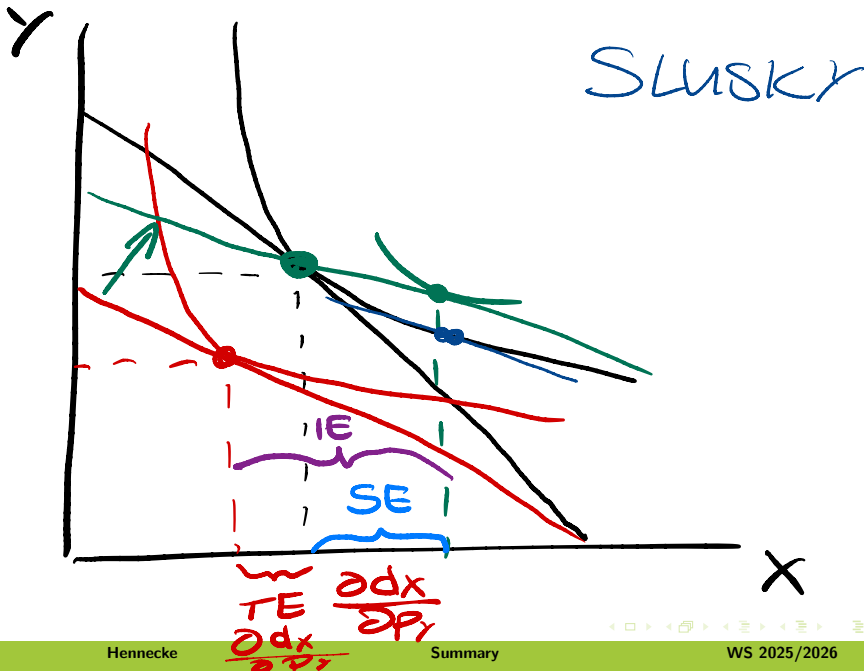
Both decompositions split a price effect into substitution and income components, but they answer **different questions**.

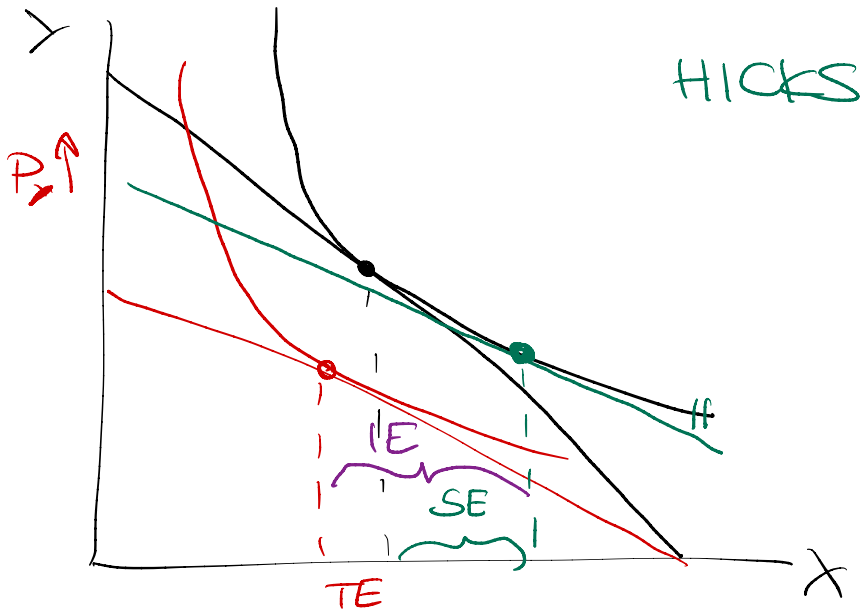
Slutsky decomposition

- Compensates income so the **original bundle remains affordable**
- Holds **purchasing power** constant
- Natural for thinking about **actual consumer responses**

Hicksian decomposition

- Compensates income so the **original utility level is maintained**
- Holds **utility** constant
- Natural for **welfare analysis**





From Slutsky Equation to Slutsky Decomposition

Recall the Slutsky equation:

$$\underbrace{\frac{\partial d_x}{\partial p_x}}_{\text{total}} = \underbrace{\frac{\partial h_x}{\partial p_x}}_{\text{SE}} - \underbrace{x \frac{\partial d_x}{\partial I}}_{\text{IE}}$$

Where is “purchasing power held constant”?

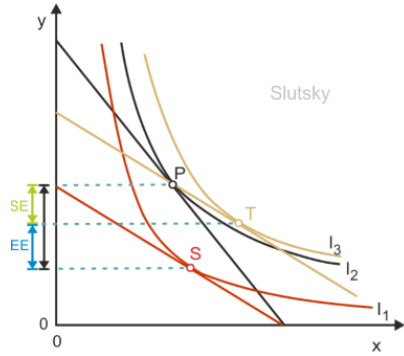
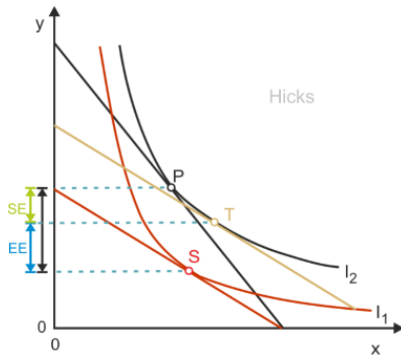
- When p_x increases by dp_x , the cost of the *original bundle* rises by

$$dI = x dp_x$$

- The term $x \frac{\partial d_x}{\partial I}$ adjusts demand for exactly this income change
- Therefore, the substitution effect is evaluated **holding the original bundle affordable**

Interpretation: Slutsky compensation restores the ability to buy the original bundle — not the original utility level.

Slutsky vs. Hicks: What exactly is held constant?



Application I: Compensation Payments

- Empirical Study: **Waldfoegel, J. (1993)** “Deadweight Loss of Christmas”
 - Empirical questions: Do (in-kind) gifts destroy welfare?
 - **Idea:** Consumers cannot be made worse off by a free monetary transfer compared to in-kind transfers (Carte Blanche Principle)
 - Slutsky vs. Hicks: Constant purchasing power $\neq$ constant utility
 - Consumers know their preferences / utility functions best, policymakers can only hold purchasing power (not utility) constant $\rightarrow$ overcompensation

Application II: Classification of goods

- **Normal:** $\frac{\partial x_i}{\partial I} > 0$ (IE reinforces SE) $\Rightarrow \frac{\partial x_i}{\partial p_i} < 0$
- **Inferior:** $\frac{\partial x_i}{\partial I} < 0$
 - **Weakly inferior:** inferior but $|IE| < |SE| \Rightarrow \frac{\partial x_i}{\partial p_i} < 0$
 - **Giffen:** inferior but $|IE| > |SE| \Rightarrow \frac{\partial x_i}{\partial p_i} > 0$
- Empirical Study: **Jensen and Miller (2008)** “Giffen Behavior and Subsistence Consumption”
 - Empirical question: Does Giffen behavior exist in reality?
 - Finding: Giffen behavior often found in subsistence consumption

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Markets: from individuals to outcomes

Two levels:

- **Partial equilibrium:** one market, “everything else fixed” ...
- ... ignore feedback effects between markets.
- But these effects are important to assess the total impact, e.g. on welfare of policy interventions.
- **General equilibrium:** All markets simultaneously → Interactions matters!

Prices coordinate decentralized choices. → Gains from trade!

Walrasian Equilibrium: Intuition via Excess Demand

Think of prices as **signals** that coordinate individual decisions.

Excess demand for a good

- Difference between how much consumers want to buy and how much is available
- *Positive excess demand*: shortages $\Rightarrow$ upward pressure on prices
- *Negative excess demand*: surpluses $\Rightarrow$ downward pressure on prices

Walrasian (competitive) equilibrium

- A vector of prices such that **no market has excess demand**
- Given these prices, all consumers choose optimally
- **All** markets clear *simultaneously*

Key idea: Equilibrium prices are those at which individual plans are mutually consistent.

Welfare theorems: the core take-away

Under **perfect market conditions** (competition/no market power, no externalities, full info, no transaction costs):

- **1st Welfare Theorem:** Competitive equilibrium is Pareto efficient
- **2nd Welfare Theorem:** Any Pareto efficient allocation can be reached after redistribution of endowments

Key implication: separation of

Efficiency (markets) vs. Equity (via redistribution)

Application: Market Failure

- If one of the “perfect market” conditions fails:
 - externalities
 - transaction costs
 - market power
 - asymmetric information and information frictions
 - behavioral biases
- *Then competitive outcomes need not be efficient — and policy can improve welfare.*
- **Empirical study: Jensen (2007)** “The digital provide: Information (technology), market performance, and welfare in the south Indian fisheries sector”
 - Illustrating the role of information (through mobile phone availability) for the functioning of markets

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Decision under risk and uncertainty

- We have to take many decisions in situations in which we don't know the outcome of the different alternatives (because outcome depends on unknown future).
- These decisions cannot be modeled with consumer theory alone because we somehow have to incorporate (preferences for) uncertainty/risk.
- This is particularly true as many people are not indifferent towards risk.
- Distinction:
 - Uncertainty (probabilities unknown)
 - Risk (known probabilities of possible future states of the world)
- Standard approach for decisions under risk: expected utility theory.

Expected Utility: the benchmark model

Lotteries L over outcomes x with probabilities π .

$$EU(L) = \sum_j p_j u_j(x)$$

Foundation (vNM):

- continuity and independence $\Rightarrow$ EU representation
- only the **reduced lottery** matters (consequentialism)

Powerful model — but with strong behavioral assumptions.

Risk preferences: curvature and what it means

- **Risk averse:** concave $u(\cdot)$
 - Jensen: $EU(L) < U(E(L))$
 - Arrow–Pratt measures: ARA / RRA quantify “how concave”
- **Risk neutral:** linear $u(\cdot)$
 - Jensen: $EU(L) = U(E(L))$
- **Risk loving:** convex $u(\cdot)$
 - Jensen: $EU(L) > U(E(L))$

Empirical Study: Dohmen et al. (2011) “Individual Risk Attitudes: Measurement, Determinants, and Behavioral Consequences”

- Different ways of measuring individual risk attitudes

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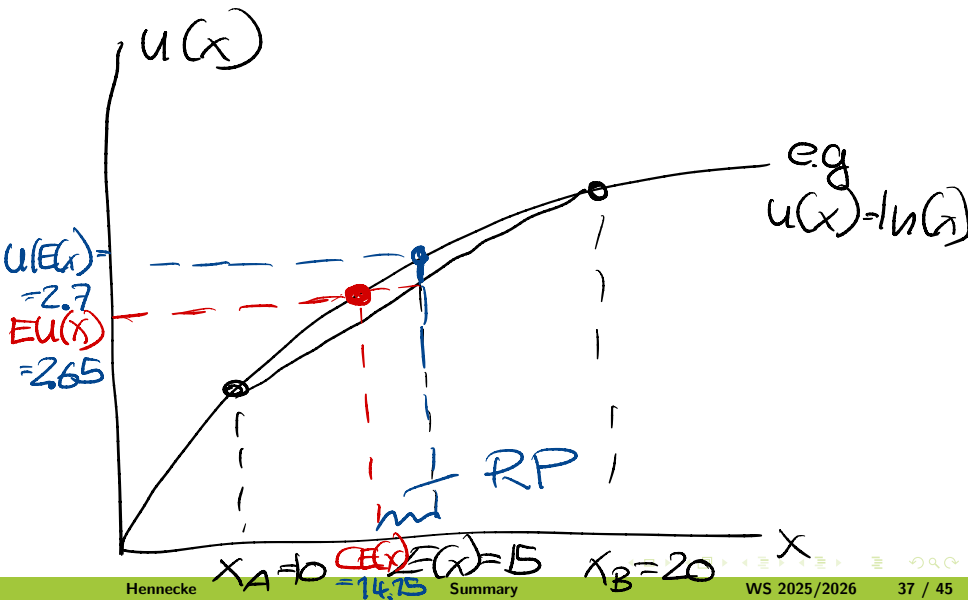
Critical Concepts - Certainty Equivalent

- Risk preferences (w.r.t. a specific lottery) are reflected in the

Certainty equivalent

c of a lottery $L = (p_1, p_2)$ with outcomes x_1 and x_2 is the amount of money that - when received with certainty - gives an individual the same expected utility as the lottery. c is implicitly defined by $U(c) = p_1 U(x_1) + p_2 U(x_2)$.

Risk Aversion



Insurance: why it can increase welfare

With fair insurance contracts:

- risk-averse consumers prefer to smooth consumption across states
- optimal choice: (often) full insurance

In theory: Markets for interdependent risks

- **Risk pooling** (law of large numbers) or **risk transfer** (to those with lower marginal disutility)

In practice: adverse selection, moral hazard, correlated risks.

Implicit markets for risk & Value of Statistical Life (VSL)

Even without explicit insurance markets, risk is traded:

- wage premia for risky jobs (compensating differentials)
- safety products / time vs. risk trade-offs

These trade-offs reveal willingness to pay for safety:

$$\text{money} \leftrightarrow \text{mortality risk} \Rightarrow \text{VSL}$$

- Empirical Study: **Ashenfelter and Greenstone (2004)** “Using Mandated Speed Limits to Measure the Value of a Statistical Life”
 - Decisions under risk can be inferred to learn about individual preferences (“revealed preferences”)
 - Here used to inform the question of how much a (statistical) life is worth
- **Policy relevance:** regulation may improve welfare when implicit markets fail (info frictions, biases, externalities).

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Behavioral Economics

- Standard theory is useful as a “normative” theory but ...
- ... makes predictions which do not hold as a “positive”/descriptive theory, i.e. they cannot describe how humans make decisions in the real world
- Predictions deviate, e.g. as people are Humans, not Econs, they often
 - are not capable of understanding/thinking through the full consequences of their decisions
 - lack self-control to stick to their plans
 - fear losses more than they like gains (loss aversion)
 - care for others, not just about themselves
- Many of these deviations were documented first with **experimental economics**
- **Behavioral economics** tries to bring the Human back into economics

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Closing message

- **Key takeaway:** Assumptions matter! - Deviations from assumptions (about individual preferences or markets) make models fail and lead to pathways for policy interventions.

Think like an economist:

Be explicit about assumptions,
be clear about mechanisms,
and be humble about predictions.

Hints for the Exam

- Similar format to mack exam / last year's exams
- Single-choice questions (20 points), Calculation (20 points), Empirical Study (20 points)
- 60 minutes - 60 points
- Non-programmable pocket calculator allowed
- Advice:
 - Give short and concise answers
 - Write down your calculations (to allow us to take "follow-up" errors into account)
 - For the empirical part, you do not have to read the study first - read the questions and search for what you need to answer them in the text!

Questions?

